Ctrl+Alt+Decaf: Project Charter & Scope

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Project Overview

Ctrl+Alt+Decaf is a digital ordering system for the campus coffee shop that replaces handwritten orders on paper cups. Customers can order ahead on their phones and choose a pick-up time that fits their schedule. Baristas sign in quickly and take orders on a consistent screen, and see the queue so they can mark drinks as done. Managers control the menu, daily deals, discounts, and refunds on in-person payments recorded in the system, and can review the orders each employee enters.

Core Goals

- Reduce wait times with mobile ordering and customer chosen pick-up time.
- Keep item availability accurate in real time so customers are not told “sorry, we’re out” after ordering.
- Give baristas a simple order queue so they can see what to make and mark drinks as done.
- Give managers control over the menu, daily deals, discounts, and refunds.
- Make the system fast and simple for new employees, with PIN sign-in and a consistent order screen.
- Create accountability by linking every order to the employee who entered it by name, employee ID, and time stamp.

In Scope (what we will build)

- Mobile ordering with a clear, categorized menu, drink customization, and pick-up time selection.
- Fast employee sign-in using a short PIN.
- A consistent barista order screen with an order queue and a “mark done” action.
- Real-time out of stock marking visible to staff and mobile customers.
- Manager tools for updating menu items and daily deals.
- Manager-only permissions for discounts and refunds.
- Order tracking and review by employee.

Out of Scope (what we will not build right now)

- **Delivery service or third-party integration:** requires driver management and added costs.
- **Customer loyalty/rewards program:** adds customer accounts and point tracking that can wait for a later phase.
- **Automated inventory ordering from suppliers:** requires supplier integrations beyond the current need.
- **Online payment processing:** customers pay in person at pick-up for this version. In-app payments need payment security and third-party integration, so this can wait for a later phase.

User Stories

Ten Agile user stories covering the three main users: the Customer, the Barista, and the Manager.

Customer

C1	As a customer, I want to browse a menu organized into clear categories so that I can find my drink quickly.
C2	As a customer, I want to place my order from my phone so that I can skip the line.
C3	As a customer, I want to choose my flavors, milk type, toppings, and size so that my drink is made the way I like it.
C4	As a customer, I want to choose my pick-up time so that my order is ready when it fits my schedule.

Barista

B1	As a barista, I want to sign in with a short PIN before taking a customer's order so that I can start taking orders in seconds, even during a rush.
B2	As a barista, I want to see a queue of incoming orders in the same layout every time and mark each drink as done so that I can work quickly during a rush, and my team knows which drinks are finished.
B3	As a barista, I want to mark an item out of stock so that my coworkers and mobile customers see it immediately and I do not have to apologize for taking an order we cannot make.

Manager

M1	As a manager, I want to update menu items and daily deals from one screen so that prices and offerings are ready the moment we open.
M2	As a manager, I want to restrict discounts and refunds to manager-level accounts so that staff cannot issue them without approval.
M3	As a manager, I want every order tied to the employee who entered it so that I can review orders and catch possible mistakes or unauthorized changes.

Acceptance Criteria

Acceptance criteria are written for the two most important stories: B1 (PIN sign-in), because every order depends on it, and B3 (out-of-stock marking), because it protects customers from ordering items we cannot make.

Story B1: Sign-in PIN

As a barista, I want to sign in with a short PIN before taking a customer's order so that I can start taking orders in seconds, even during a rush.

Acceptance criterion (written in the Given-When-Then format)
Given a barista is on the home screen, **when** they enter their valid 4-digit PIN, **then** the order screen opens within 3 seconds.
Given a barista is on the home screen, **when** they enter an invalid PIN, **then** access is denied and the message "Incorrect PIN. Please try again." appears.
Given a barista has entered an invalid PIN twice in a row, **when** they enter a third invalid PIN, **then** that employee's sign-in is locked for 30 seconds and the message "Too many attempts. Try again in 30 seconds." appears.
Given an employee's sign-in is locked, **when** 30 seconds have passed, **then** the employee can attempt to sign in again.
Given a barista has signed in, **when** they place an order, **then** the order record shows the employee's name, employee ID, time stamp, and order description.

Story B3: Out-of-stock marking

As a barista, I want to mark an item out of stock so that my coworkers and mobile customers see it immediately and I do not have to apologize for taking an order we cannot make.

Acceptance criterion (written in the Given-When-Then format)
Given an item is in stock, **when** a barista marks it out of stock, **then** it is greyed out on the order screen and the mobile menu within 5 seconds.
Given an item is out of stock, **when** a customer tries to add it to an order, **then** the app shows the message "Sorry, this item is currently unavailable." and the item is not added.
Given a customer already has the item in their mobile cart, **when** a barista marks it out of stock and the customer tries to submit the order, **then** the customer sees the "unavailable" message and must remove the item before submitting.
Given one barista marks an item out of stock, **when** another signed-in barista views their order screen, **then** the item shows as out of stock.
Given an item is out of stock, **when** a barista marks it back in stock, **then** it becomes orderable again within 5 seconds.

System Modeling: UML Use Case Diagram

The diagram shows how the three actors (stick figures) interact with the system (boundary box) through use cases (ovals). Each use case traces back to at least one user story.

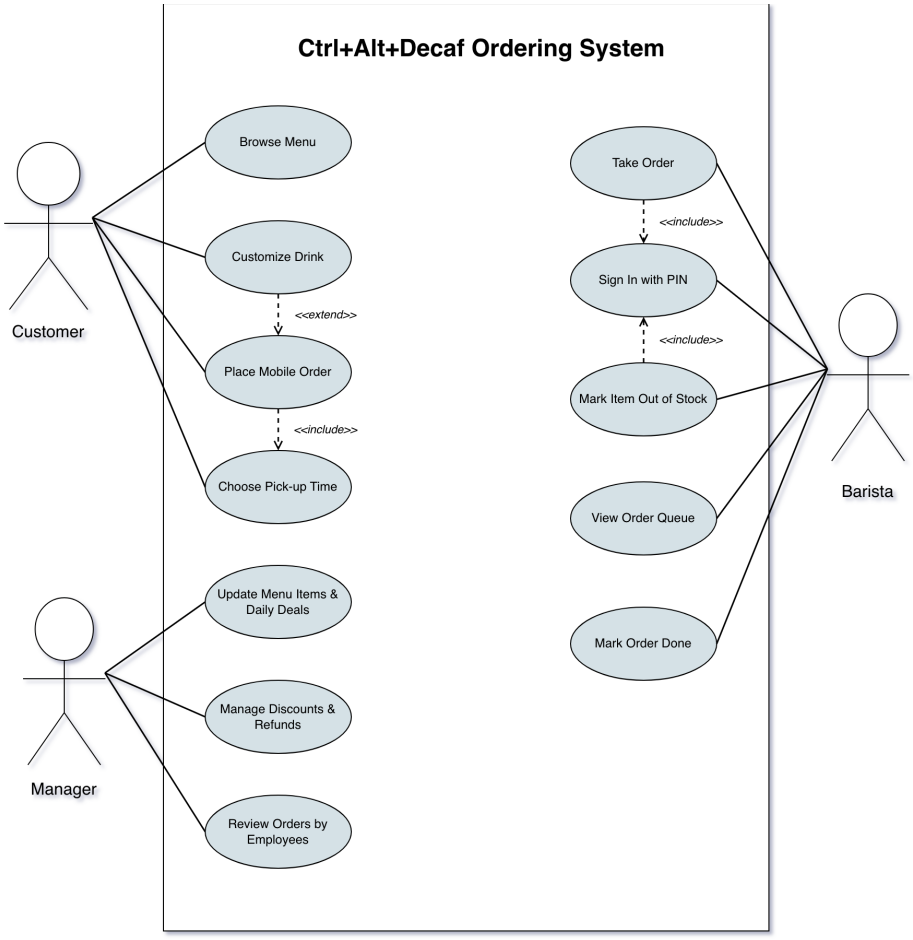


Figure 1. Use case diagram for the Ctrl+Alt+Decaf ordering system. A solid line means the person uses that feature. A dashed «include» arrow means the feature it starts from always uses the one at the arrowhead. A dashed «extend» arrow means the feature it starts from is an optional add-on to the one at the arrowhead.

Actor	Use case	Story
Customer	Browse Menu	C1
Customer	Place Mobile Order	C2
Customer	Customize Drink	C3
Customer	Choose Pick-up Time	C4
Barista	Sign In with PIN	B1
Barista	Take Order	B1
Barista	View Order Queue	B2
Barista	Mark Order Done	B2
Barista	Mark Item Out of Stock	B3
Manager	Update Menu Items & Daily Deals	M1
Manager	Manage Discounts & Refunds	M2
Manager	Review Orders by Employees	M3